

UP Board 2020

Grade 12

Chemistry

Time: 3 Hour 15 Minutes

Max. Marks: 70

Note: First 15 minutes are allotted for the candidates to read the question paper.

Instruction :

- i) All questions are compulsory. Marks allotted to each question are given in the margin.
- ii) In numerical questions, give all the steps of calculation.
- iii) Give relevant answers to the questions.
- iv) Give chemical equations wherever necessary.

1. Four alternatives are given in each part of this question. Select the correct alternative and write it in your answer-book :

- a) The unit of rate and rate constant is same for which order of reaction ?
 - i) Zero
 - ii) First
 - iii) Second
 - iv) Third

b) In which of the following lowering of vapour pressure would be maximum ?

- i) 0.1 M BaCl₂
- ii) 0.1 M Glucose
- iii) 0.2 M Urea
- iv) 0.1 M NaCl.

1

c) In a crystal system $a = b = c$ and $\alpha = \beta = \gamma \neq 90^\circ$, this system is

- i) Tetragonal
- ii) Hexagonal
- iii) Rhombohedral
- iv) Monoclinic.

1

d) In a chemical reaction, the effect of catalyst is to change

- i) Activation energy
- ii) Equilibrium concentration
- iii) Heat of reaction
- iv) Final product.

1

e) In the presence of light, chloroform oxidises to produce

- i) Carbon tetrachloride
- ii) Carbonyl chloride
- iii) Dichloromethane

iv) None of these.

1

f) $(\text{CH}_3)_3\text{-N}$ is a

i) Primary amine

ii) Secondary amine

iii) Tertiary amine

iv) none of these.

1

2. a) An aqueous solution of an organic substance, in which 3 Gm of substance is dissolved in 50 gm water, has a boiling point of 100.51°C . Calculate the molecular weight of the substance. (k_b for water = $0.51^\circ\text{C molar}^{-1}$). 2

b) The standard molar conductivity Λ_m° , for NaCl, HCl and CH_3COONa are $126.4 \text{ S cm}^2 \text{ M}^{-1}$, $425.9 \text{ S cm}^2 \text{ M}^{-1}$ and $91.0 \text{ S cm}^2 \text{ M}^{-1}$ respectively. Calculate Λ for CH_3COOH . 2

c) Explain the characteristics of enzyme catalysis. 2

d) Write four important uses of noble gas, helium. 2

3. a) What are isotonic solutions ? Explain with example. 1 + 1

b) What are meant by n-type and p-type semiconductors ?

Write their uses.

1 + 1

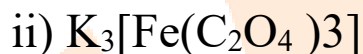
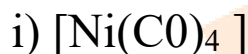
c) Write a short note on secondary batteries. 2

d) Write short notes on the following :

i) Nernst equation

- ii) Tyndall effect. 1 + 1
4. a) What are transition elements ? Explain their magnetic and catalytic properties. 1 + 1 + 1
- b) Write the uses of freon. What is its impact on environment ? 1 + 2
- c) Write the chemical equation for the preparation of methylamine by Schmidt reaction. Write chemical equations for the reactions taking place on heating (i) Methylamine and (ii) Aniline with chloroform in the presence of alcoholic KOH. 1+1+1
- d) What are addition and condensation polymers ? Give two examples for each. Write the chemical equation for the formation of Bakelite and also two uses of Bakelite. 1+1+ $\frac{1}{2}$ + $\frac{1}{2}$
5. a) Explain the meaning of rate of reaction with examples. A first order reaction is 20% complete in 10 minutes. Find the time required for 75% completion of the reaction. $\frac{1}{2}$ + 3 $\frac{1}{2}$
- b) Explain the process of extraction of Aluminium. Give a labelled diagram and also give chemical equations for the reactions taking place. 2 + 1 + 1

c) What is the reason for optical isomerism in co-ordination compounds ? Explain by giving one example. Give IUPAC names of the following compounds :



2 + 1 + 1

d) What are soaps ? Write the method of its preparation in short. Why does soap not work in hard water ? What is the primary structure of protein ? Explain with example

2 + 1 + 1

6: a) Describe the methods for preparation of phenol from

(i) Diazonium salts and (ii) Cumene. Write chemical equations for Kolbe reaction and Reimer-Tiemann reaction of phenol.

1 ½ + 1½ + 1 + 1

OR

Write two methods for preparation and three chemical properties of Alcohol and give chemical equations also.

1 + 1 + 1 + 1 + 1

b) What happens when (Write chemical equations only)

i) Acetone reacts with iodine in presence of NaOH ?

ii) Fehling solution is mixed in Formic acid ?

iii) Benzoic acid is heated with ammonia ?

iv) Tollen's reagent is mixed with acetaldehyde ?

v) Acetic acid is heated with CaO ? **1 + 1 + 1 + 1 + 1**

OR

How will you obtain (Give chemical equations only)

i) Calcium formate from formic acid ?

ii) Urotropine from formaldehyde ?

iii) Acetyl chloride from acetic acid ?

iv) Acetone hydrazone from acetone ?

v) m-bromobenzoic acid from benzoic acid ?

1 + 1 + 1 + 1 + 1

7 a) Explain Haber process for manufacture of ammonia with the help of flow chart and chemical equations. Write two important uses of Ammonia.

2 + 2 + 1

OR

Explain the method for preparation of sulphur dioxide giving chemical equations. Give chemical equation for the reaction of sulphur dioxide with the following :

i) Sodium hydroxide solution

ii) Chlorine gas

iii) Oxygen gas.

2 + 1 + 1 + 1

b) What is the importance of proteins in our food ? What are the main sources of protein ? What do you know about secondary structure of proteins ? What is denaturation of protein ? **2 + 1 + 2**

OR

What do you know about monosaccharides, disaccharides and polysaccharides ? Explain with examples. How can glucose and starch be differentiated ? **3 + 2**